

CSE 311:

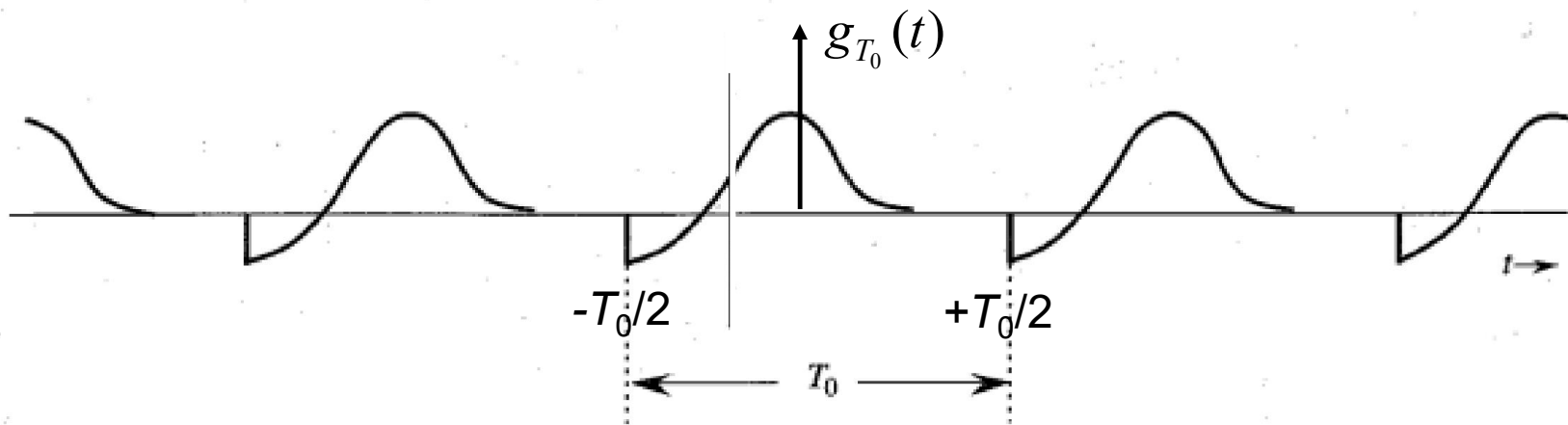
Data Communication

Instructor:
Dr. Md. Monirul Islam

Analysis of Signals and Systems

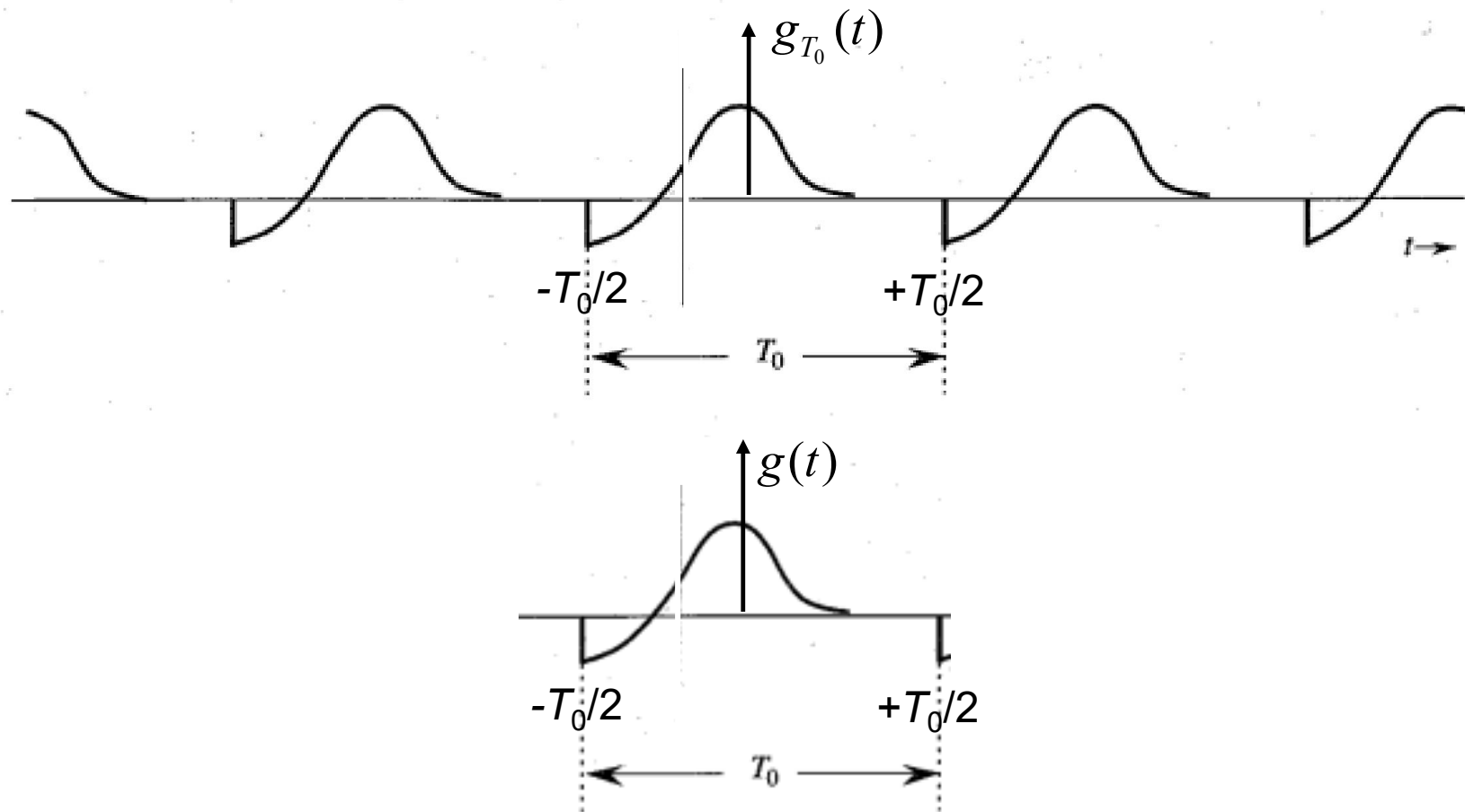
Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



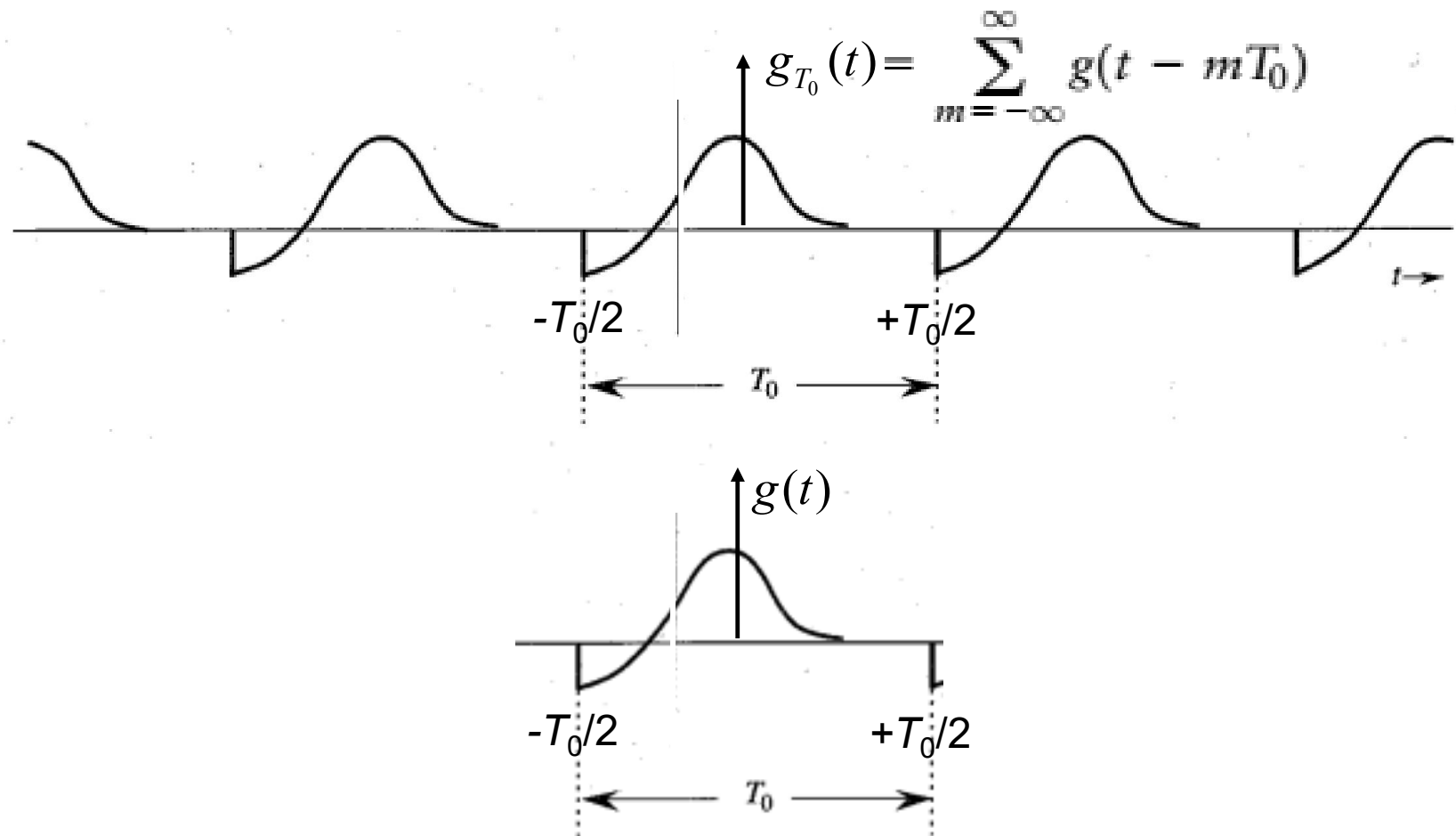
Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



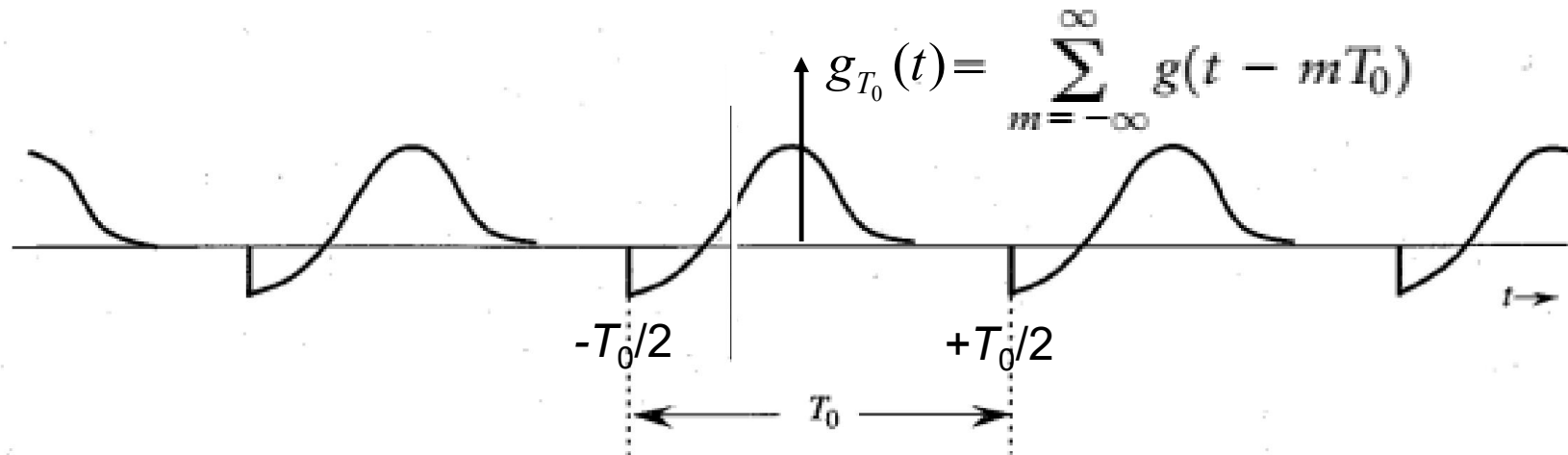
Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$

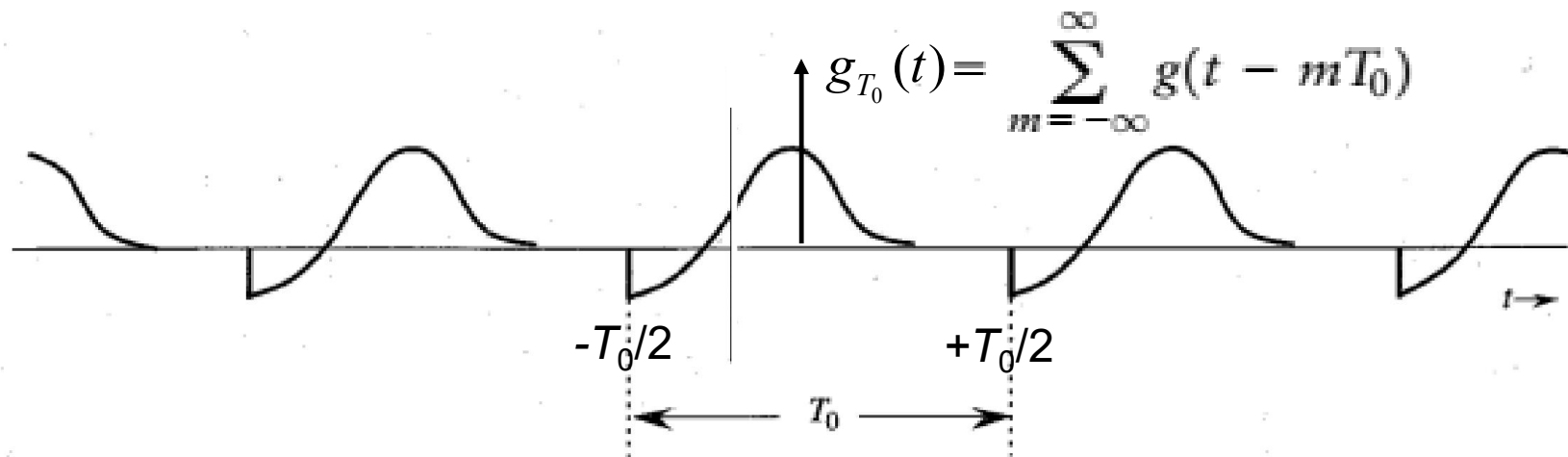


$$g(t) = \begin{cases} g_{T_0}(t), & -\frac{T_0}{2} \leq t \leq \frac{T_0}{2} \\ 0, & \text{elsewhere} \end{cases}$$

The graph shows the function $g(t)$ on a horizontal time axis. It is a single period of the periodic signal $g_{T_0}(t)$, centered at the origin, with a period T_0 marked by vertical dashed lines at $-T_0/2$ and $+T_0/2$, and a double-headed arrow below the axis labeled T_0 .

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$

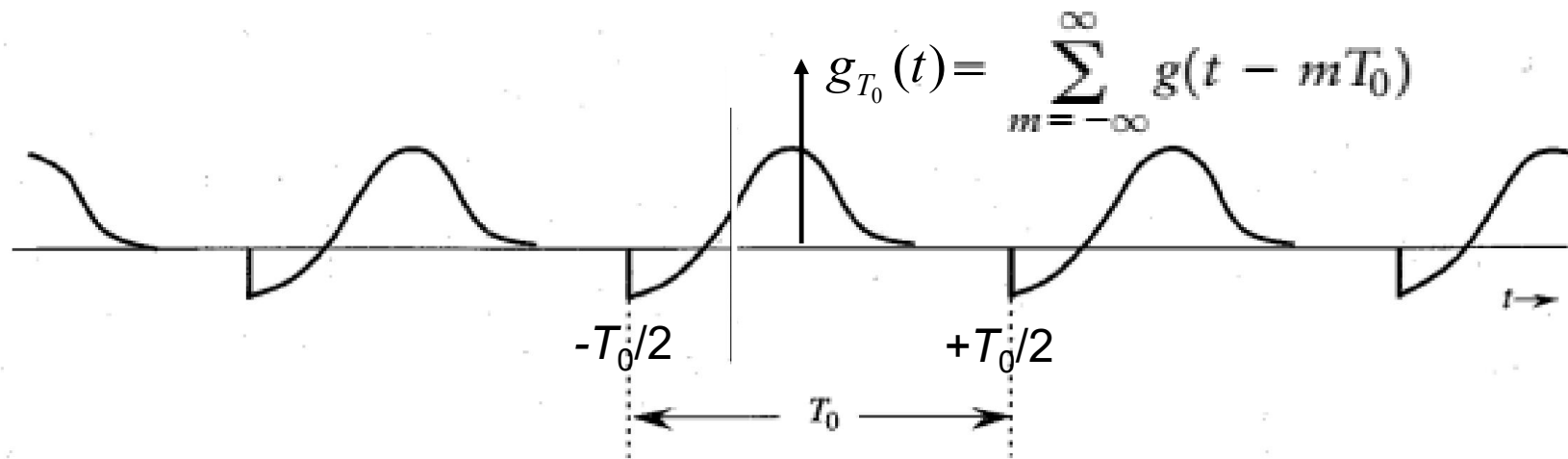


$$g(t) = \begin{cases} g_{T_0}(t), & -\frac{T_0}{2} \leq t \leq \frac{T_0}{2} \\ 0, & \text{elsewhere} \end{cases}$$

Assume that
 $\mathfrak{F}[g(t)] = G(f)$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



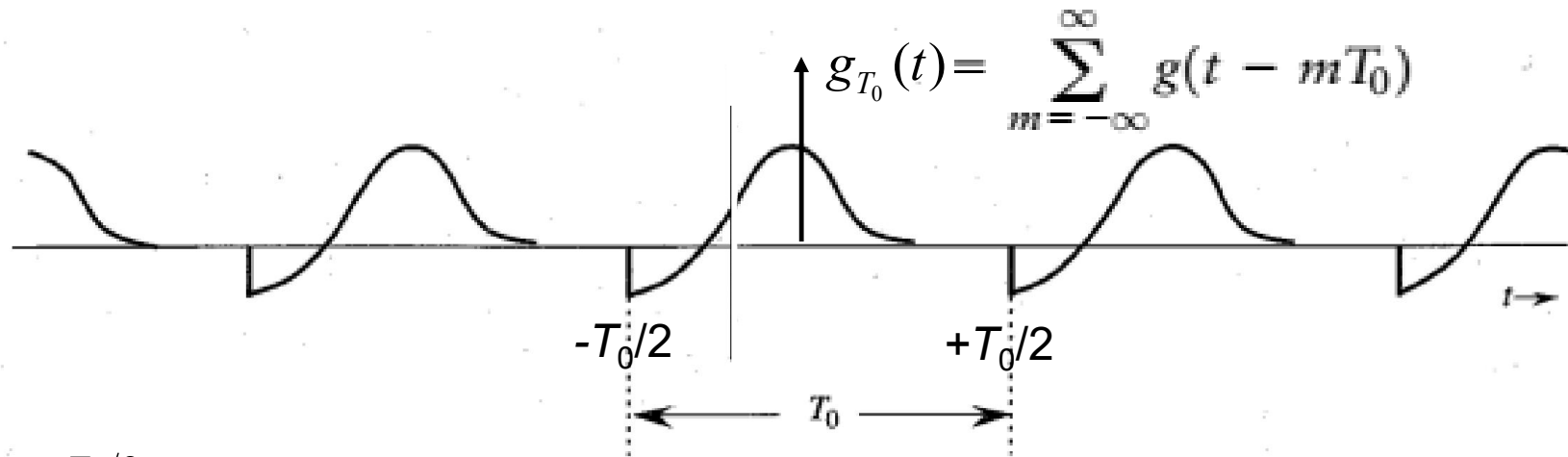
As it is periodic,

$$g_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} c_n \exp(j2\pi n f_0 t)$$

where, $c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt$ and $f_0 = \frac{1}{T_0}$

Fourier Transform of Some Useful Functions

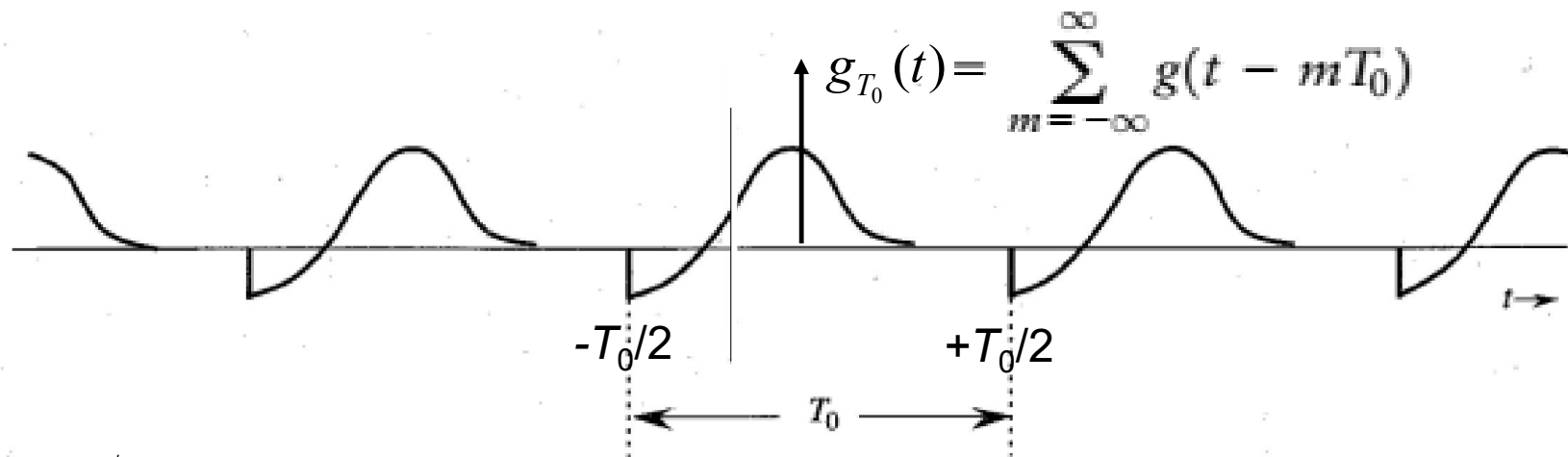
Fourier Transform of a periodic signal $g_{T_0}(t)$

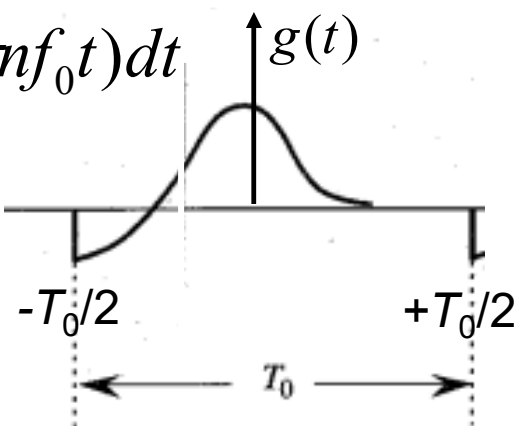


$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$

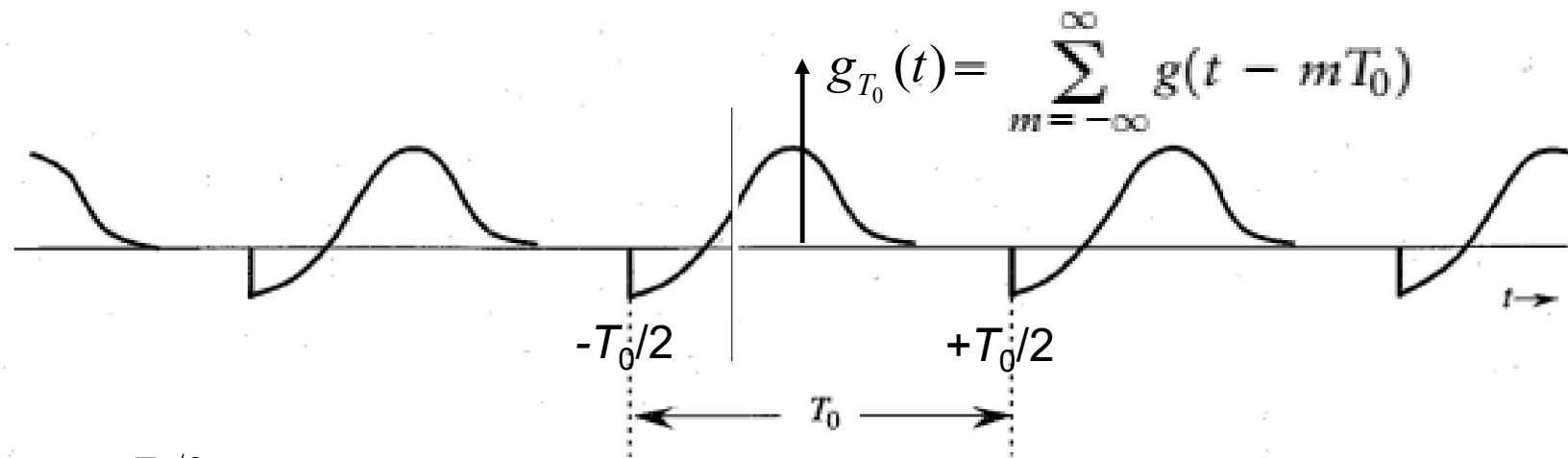


$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt$$


The graph shows the function $g(t)$ plotted against time t . The function $g(t)$ is a periodic waveform with a period T_0 . The period T_0 is indicated by a double-headed arrow between two consecutive peaks. The period T_0 is also indicated by a double-headed arrow between two consecutive peaks. The period T_0 is also indicated by a double-headed arrow between two consecutive peaks.

Fourier Transform of Some Useful Functions

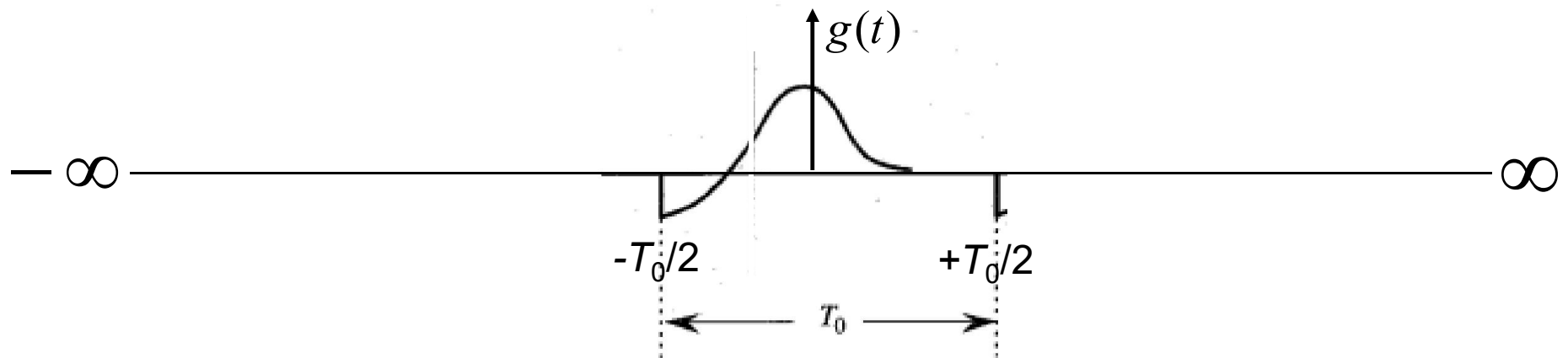
Fourier Transform of a periodic signal $g_{T_0}(t)$



$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g(t) \exp(-j2\pi n f_0 t) dt$$

Fourier Transform of Some Useful Functions

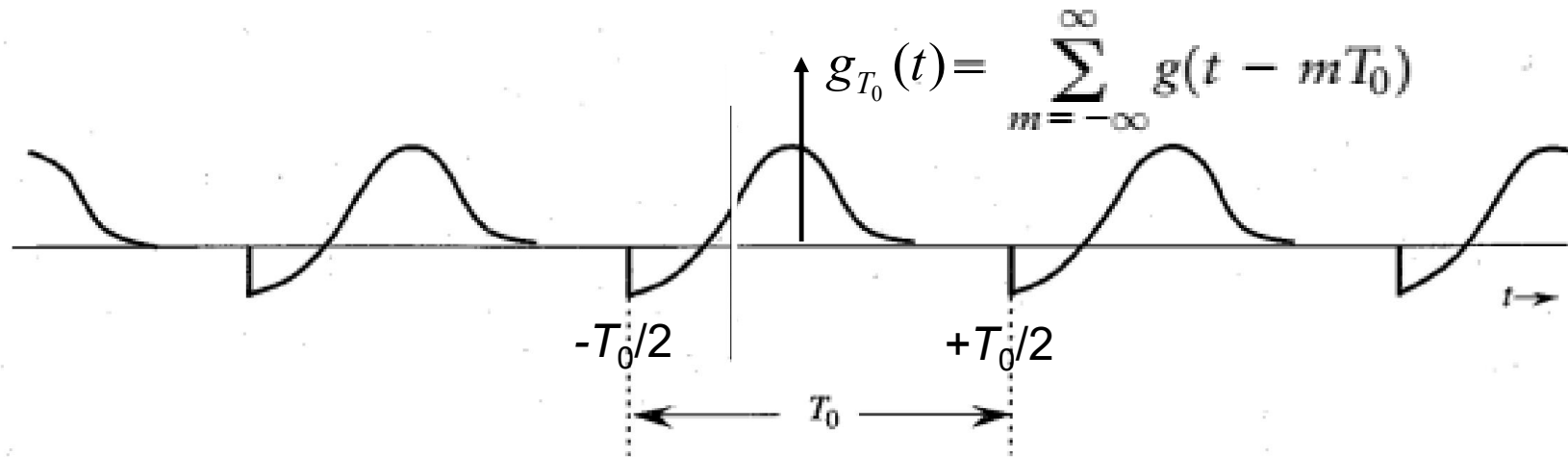
Fourier Transform of a periodic signal $g_{T_0}(t)$



$$\begin{aligned} c_n &= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} g(t) \exp(-j2\pi n f_0 t) dt \\ &= f_0 \int_{-\infty}^{\infty} g(t) \exp(-j2\pi n f_0 t) dt \\ &= f_0 G(n f_0) \end{aligned}$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



$$c_n = \frac{1}{T_0} \sum_{-T_0/2}^{T_0/2} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt$$

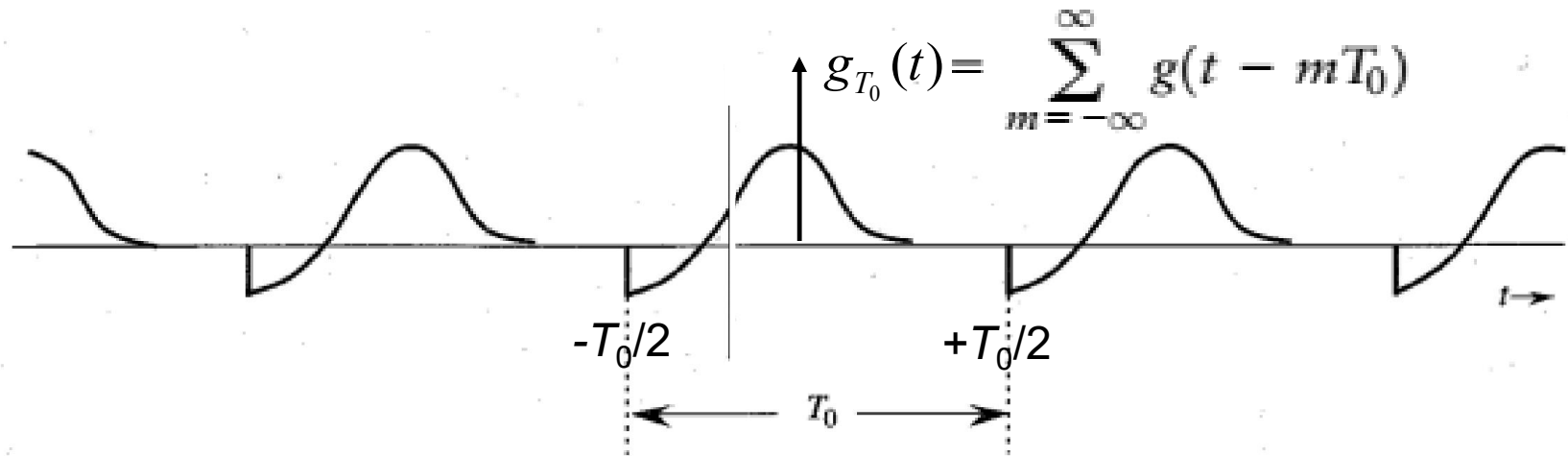
$$= f_0 \int_{-\infty}^{\infty} g(t) \exp(-j2\pi n f_0 t) dt$$

$$= f_0 G(nf_0)$$

where $G(nf_0)$ is FT of $g(t)$ evaluated at $f = nf_0$

Fourier Transform of Some Useful Functions

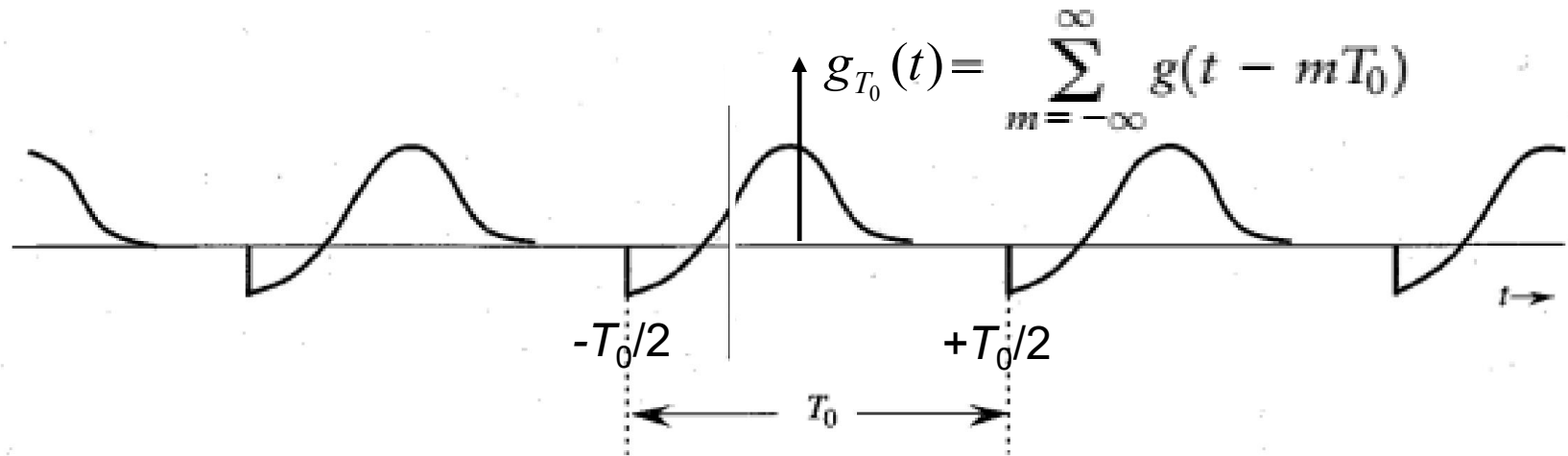
Fourier Transform of a periodic signal $g_{T_0}(t)$



$$g_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} c_n \exp(j2\pi n f_0 t) \quad \text{and we found, } c_n = f_0 G(nf_0)$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$

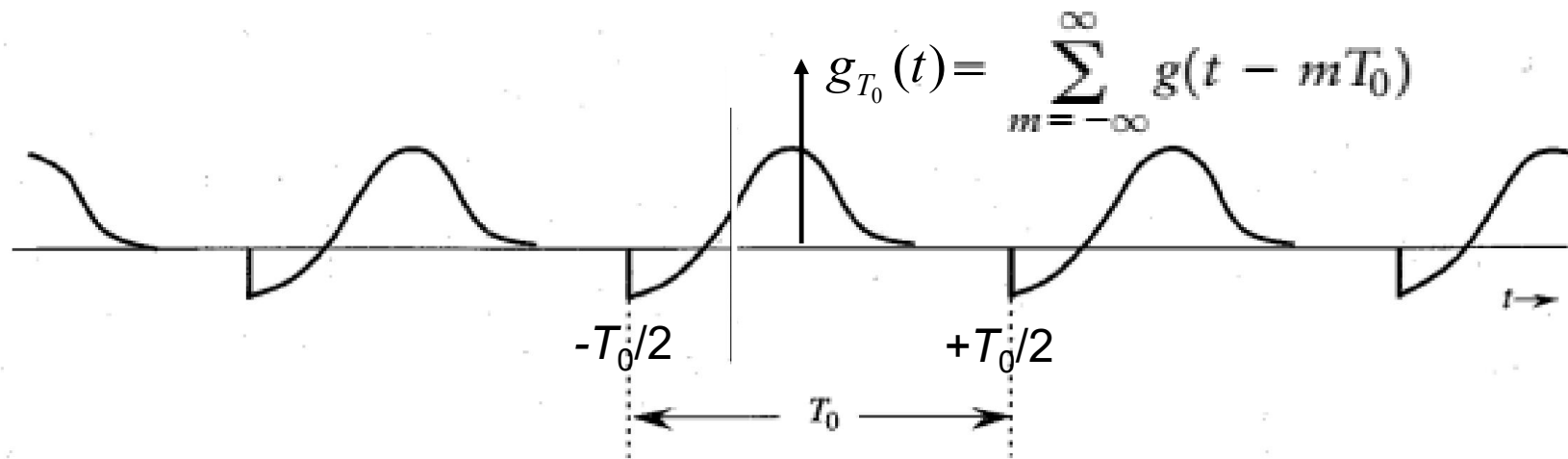


$$g_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} c_n \exp(j2\pi n f_0 t) \quad \text{and we found, } c_n = f_0 G(nf_0)$$

$$= f_0 \sum_{n=-\infty}^{\infty} G(nf_0) \exp(j2\pi n f_0 t)$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$



$$\begin{aligned}\mathfrak{T}[g_{T_0}(t)] &= \mathfrak{T}\left[f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \exp(j2\pi nf_0 t)\right] = f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \mathfrak{T}[\exp(j2\pi nf_0 t)] \\ &= f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \delta(f - nf_0)\end{aligned}$$

Fourier Transform of Some Useful Functions

Fourier Transform of a periodic signal $g_{T_0}(t)$

$$\mathfrak{F}[g_{T_0}(t)] = f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \delta(f - nf_0) \quad \mathfrak{F}[g(t)] = G(f)$$

- $g(t)$ is a single period of $g_{T_0}(t)$ but its spectrum is continuous
- Spectrum of $g_{T_0}(t)$ is discrete
- Spectrum of $g_{T_0}(t)$ consists of
 - Impulse functions occurring at integer multiples of f_0
 - each impulse function is weighted by a factor equal to the corresponding value of $G(nf_0)$

Fourier Transform of Some Useful Functions

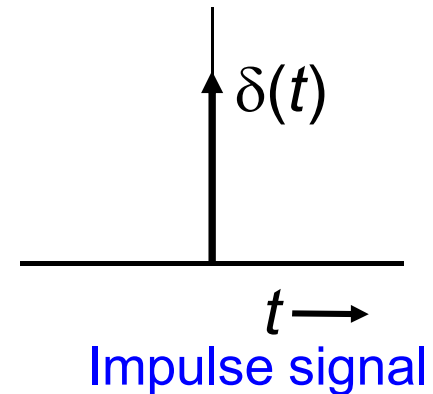
Fourier Transform of an impulse train $\delta_{T_0}(t)$

- $\delta_{T_0}(t)$ is a periodic signal with period T_0

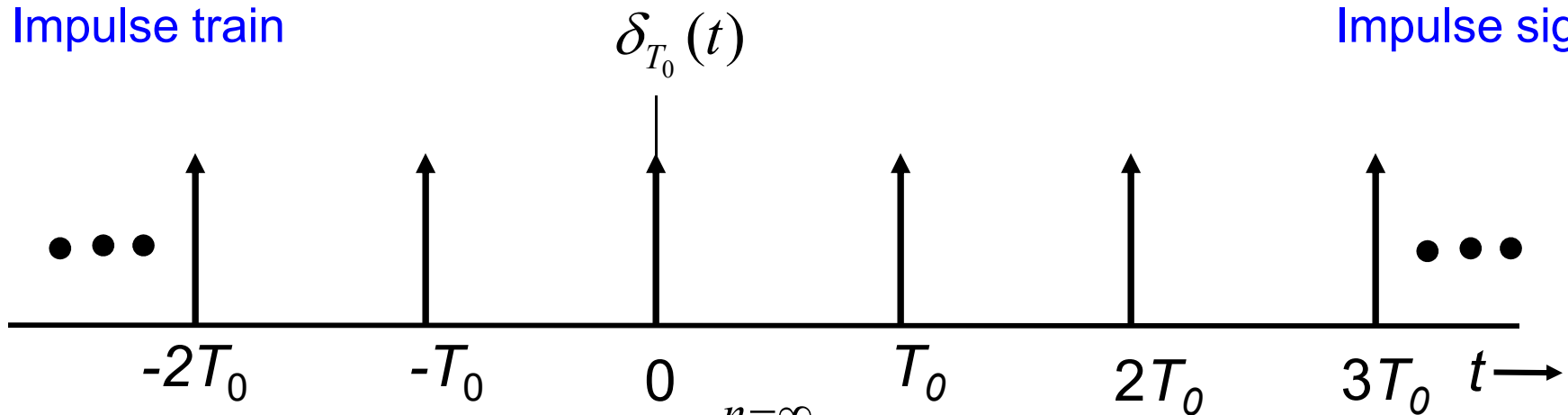
Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

- $\delta_{T_0}(t)$ is a periodic signal with period T_0



Impulse train

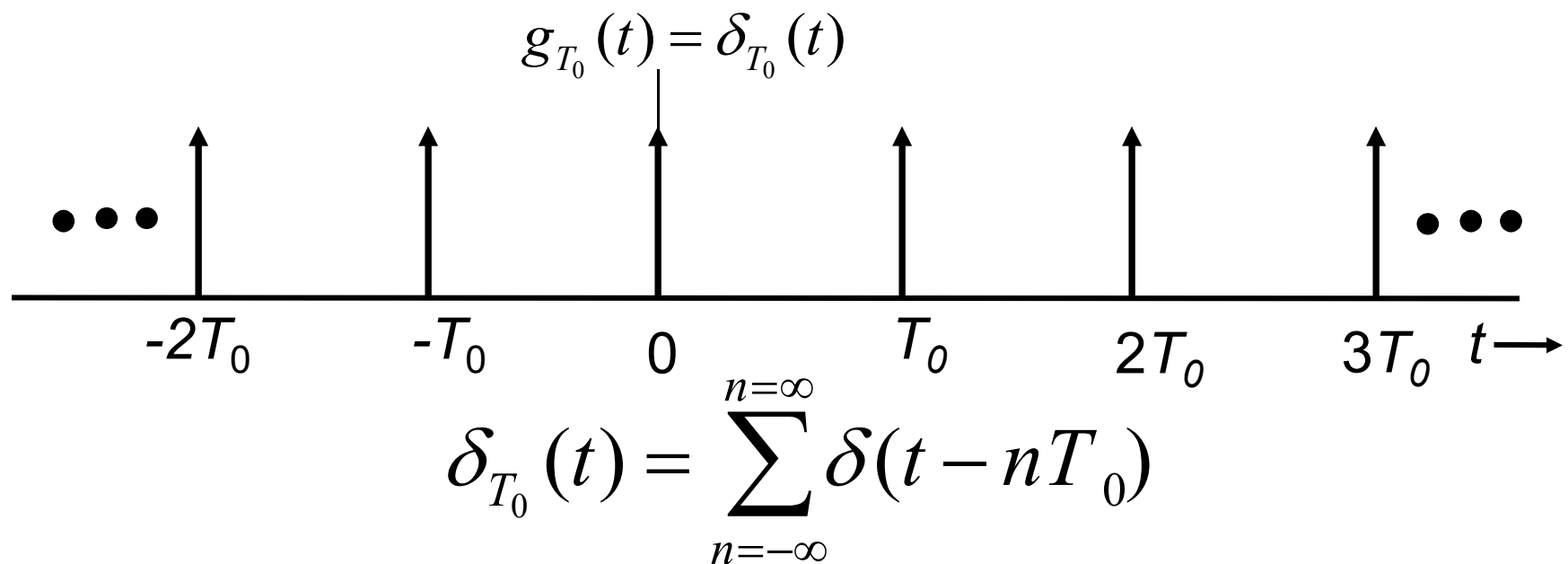


Equation of an
Impulse train

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} \delta(t - nT_0)$$

Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

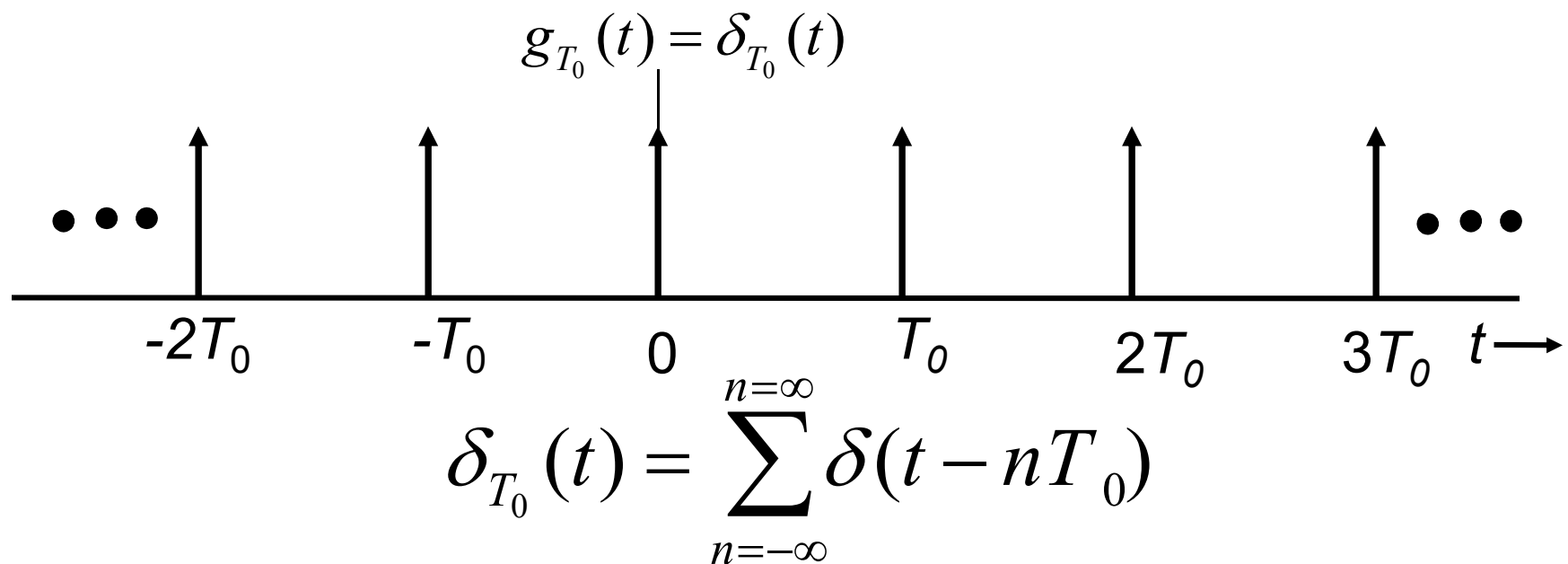


Comparing with the previous result

$$\mathfrak{F}[g_{T_0}(t)] = f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \delta(f - nf_0) \qquad \mathfrak{F}[g(t)] = G(f)$$

Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

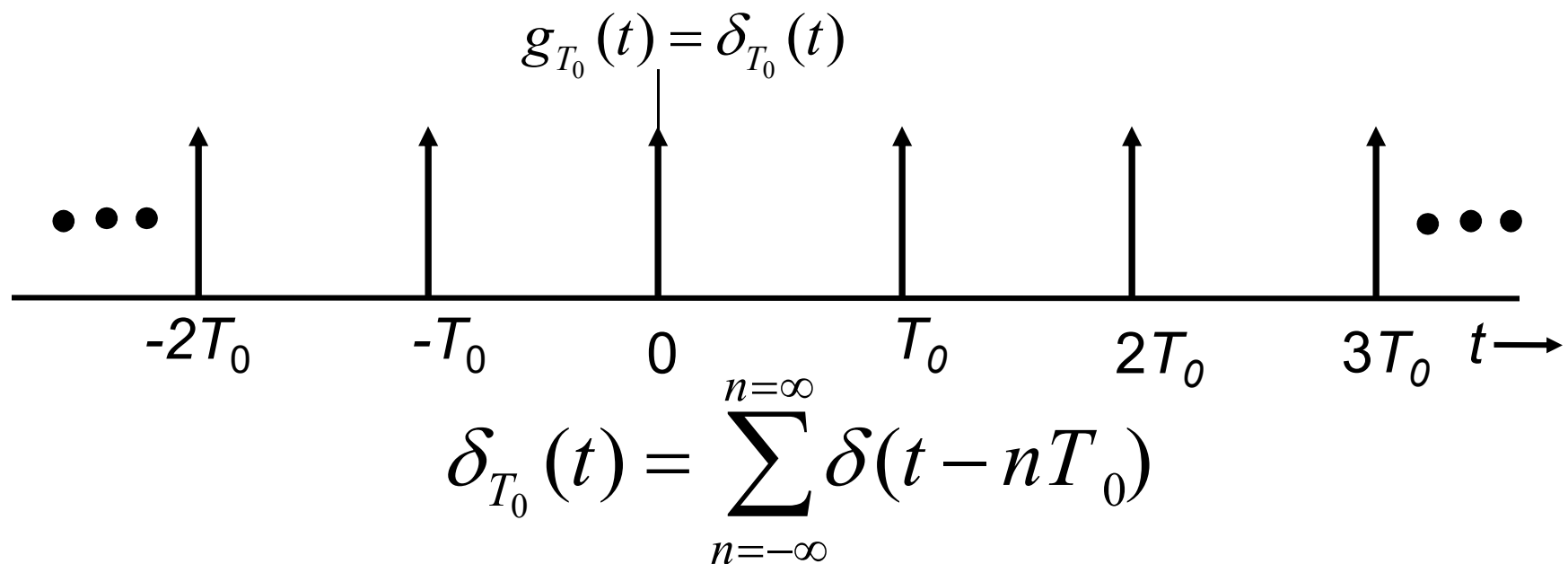


Comparing with the previous result

$$G(f) = \mathfrak{F}[g(t)] = \mathfrak{F}[\delta(t)] = 1$$

Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

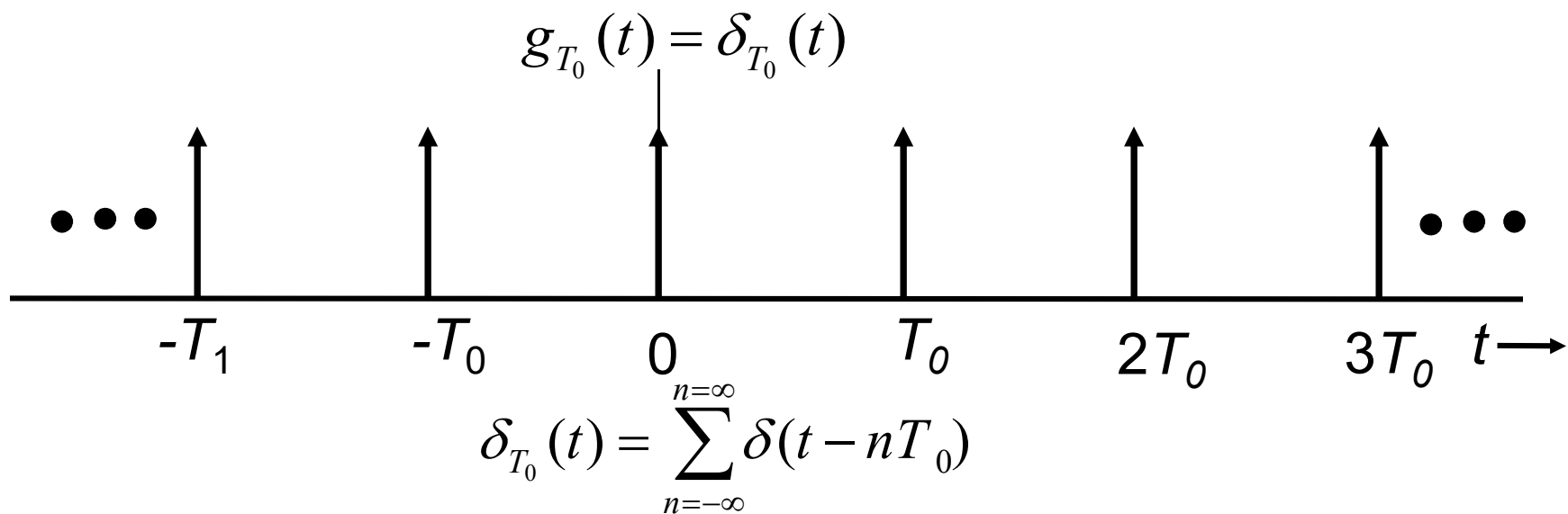


Comparing with the previous result

$$G(f) = \mathfrak{F}[g(t)] = \mathfrak{F}[\delta(t)] = 1 = G(nf_0)$$

Fourier Transform of Some Useful Functions

Fourier Transform of an impulse train $\delta_{T_0}(t)$

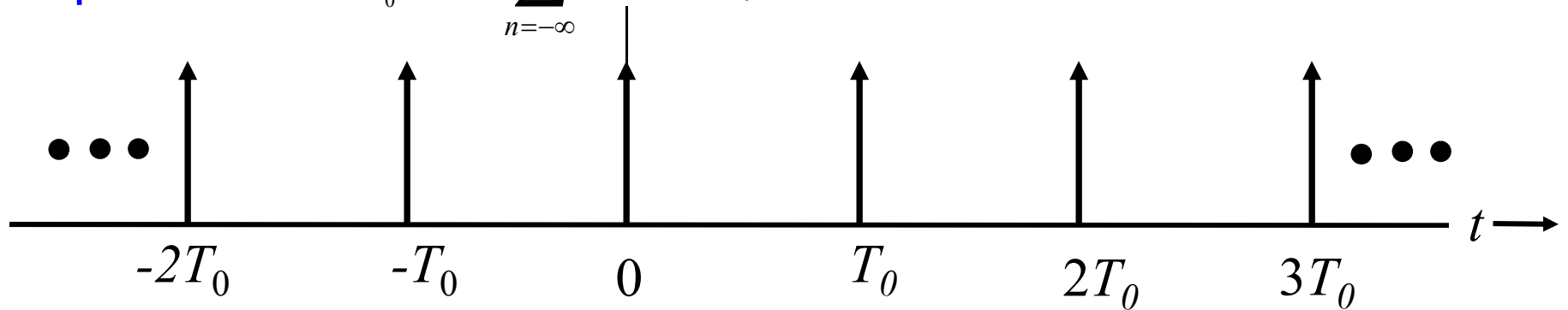


Comparing with the previous result

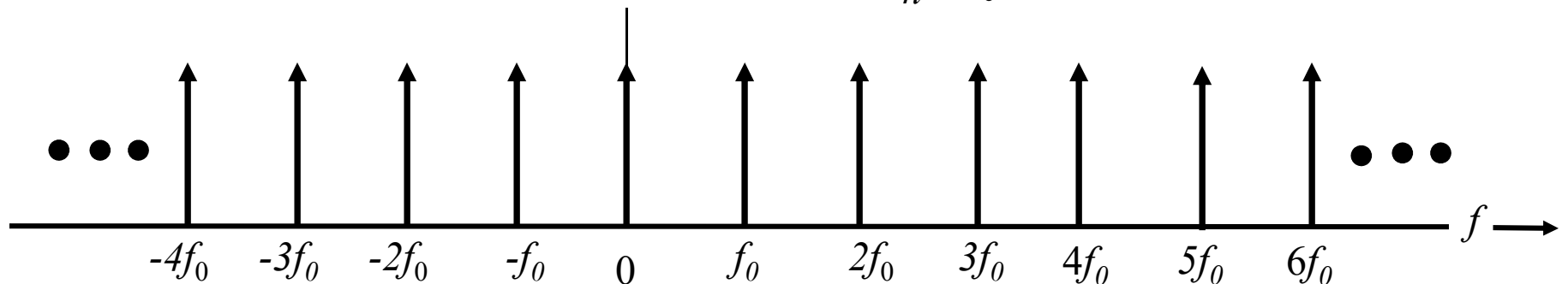
$$\mathfrak{F}[\delta_{T_0}(t)] = f_0 \sum_{n=-\infty}^{n=\infty} G(nf_0) \delta(f - nf_0) = f_0 \sum_{n=-\infty}^{n=\infty} \delta(f - nf_0)$$

Fourier Transform of Some Useful Functions

Impulse train $\delta_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} \delta(t - nT_0)$



FT of Impulse train $\mathfrak{F}[\delta_{T_0}(t)] = f_0 \sum_{n=-\infty}^{n=\infty} \delta(f - nf_0)$



Fourier Transform of Some Useful Functions

$g(t)$	$G(f)$
1 $e^{-at}u(t)$	$\frac{1}{a + j2\pi f} \quad a > 0$
2 $e^{at}u(-t)$	$\frac{1}{a - j2\pi f} \quad a > 0$
3 $e^{-a t }$	$\frac{2a}{a^2 + (2\pi f)^2} \quad a > 0$
4 $te^{-at}u(t)$	$\frac{1}{(a + j2\pi f)^2} \quad a > 0$
5 $t^n e^{-at}u(t)$	$\frac{n!}{(a + j2\pi f)^{n+1}} \quad a > 0$
6 $\delta(t)$	1
7 1	$\delta(f)$
8 $e^{j2\pi f_0 t}$	$\delta(f - f_0)$
9 $\cos 2\pi f_0 t$	$0.5 [\delta(f + f_0) + \delta(f - f_0)]$
10 $\sin 2\pi f_0 t$	$j0.5 [\delta(f + f_0) - \delta(f - f_0)]$

$g(t)$	$G(f)$
17 $\Pi\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc}(\pi f \tau)$
18 $2B \operatorname{sinc}(2\pi Bt)$	$\Pi\left(\frac{f}{2B}\right)$
19 $\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2} \operatorname{sinc}^2\left(\frac{\pi f \tau}{2}\right)$

Properties of Fourier Transform

Time-Frequency duality

Analysis equation:

$$G(f) = \int_{-\infty}^{\infty} g(t) \exp(-j2\pi ft) dt$$

Time-domain
description:
 $g(t)$

Frequency-domain
description:
 $G(f)$

Synthesis equation:

$$g(t) = \int_{-\infty}^{\infty} G(f) \exp(j2\pi ft) df$$

Properties of Fourier Transform

Time-Frequency duality

Analysis equation:

$$G(f) = \int_{-\infty}^{\infty} g(t) \exp(-j2\pi ft) dt$$

- Difference is minor
- Remarkably similar

Time-domain
description:
 $g(t)$

Frequency-domain
description:
 $G(f)$

Synthesis equation:

$$g(t) = \int_{-\infty}^{\infty} G(f) \exp(j2\pi ft) df$$

Properties of Fourier Transform

Time-Frequency duality

Analysis equation:

$$G(f) = \int_{-\infty}^{\infty} g(t) \exp(-j2\pi ft) dt$$

- Difference is minor
- Remarkably similar

Time-domain
description:
 $g(t)$

Frequency-domain
description:
 $G(f)$

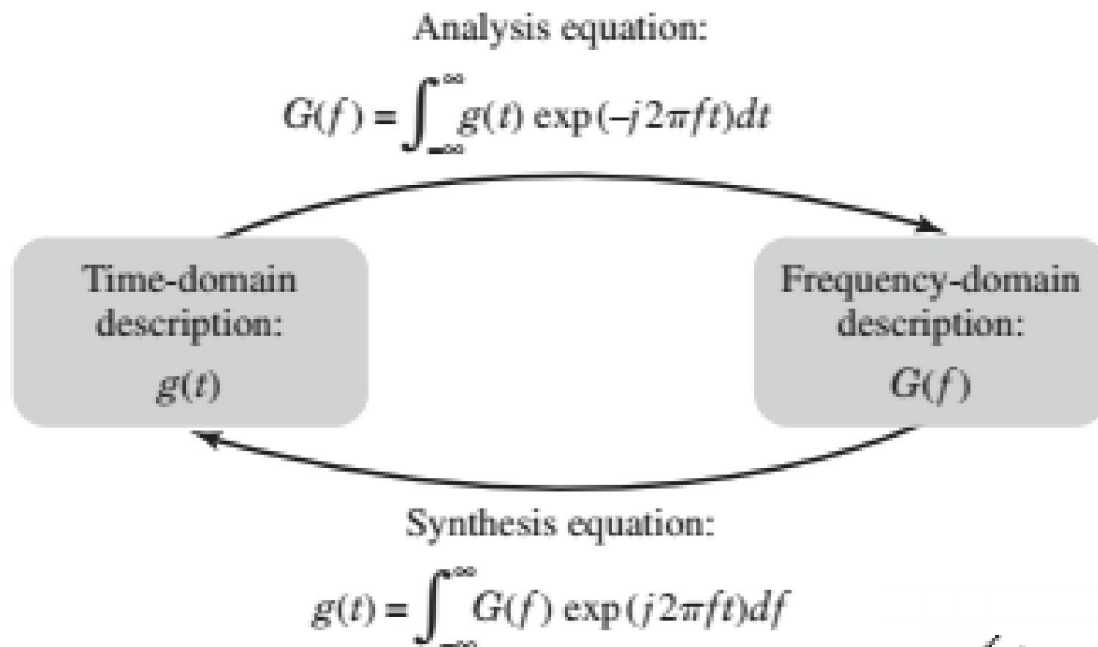
Synthesis equation:

$$g(t) = \int_{-\infty}^{\infty} G(f) \exp(j2\pi ft) df$$

- For any relationship, dual exists by interchanging roles of $g(t)$ and $G(f)$

Properties of Fourier Transform

Time-Frequency duality



Let, $g(t) \Leftrightarrow G(f)$

$$g(t - t_0) \Leftrightarrow G(f) e^{-j2\pi f t_0}$$

means

$$g(t) e^{j2\pi f_0 t} \Leftrightarrow G(f - f_0)$$

Properties of Fourier Transform

Duality property

$$\text{If } g(t) \Leftrightarrow G(f)$$

$$\text{then } G(t) \Leftrightarrow g(-f)$$

Properties of Fourier Transform

Duality property

Proof

If $g(t) \Leftrightarrow G(f)$

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

then $G(t) \Leftrightarrow g(-f)$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} G(x) e^{j2\pi xt} dx$$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} G(x) e^{j2\pi xt} dx$$

$$g(-t) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xt} dx \quad \text{Replaced } t \text{ by } -t$$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} G(x) e^{j2\pi xt} dx$$

$$g(-t) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xt} dx \quad \text{Replaced } t \text{ by } -t$$

$$g(-f) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xf} dx \quad \text{Replaced } t \text{ by } f$$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} G(x) e^{j2\pi xt} dx$$

$$g(-t) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xt} dx \quad \text{Replaced } t \text{ by } -t$$

$$g(-f) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xf} dx \quad \text{Replaced } t \text{ by } f$$

$$g(-f) = \int_{-\infty}^{\infty} G(t) e^{-j2\pi ft} dt \quad \text{Replaced } x \text{ by } t$$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} G(x) e^{j2\pi xt} dx$$

$$g(-t) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xt} dx \quad \text{Replaced } t \text{ by } -t$$

$$g(-f) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xf} dx \quad \text{Replaced } t \text{ by } f$$

$$g(-f) = \int_{-\infty}^{\infty} G(t) e^{-j2\pi ft} dt \quad \text{Replaced } x \text{ by } t$$

This means, $g(-f) = \mathfrak{I}[G(t)]$

Properties of Fourier Transform

Duality property

If $g(t) \Leftrightarrow G(f)$

then $G(t) \Leftrightarrow g(-f)$

Proof

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df = \int_{-\infty}^{\infty} G(x) e^{j2\pi xt} dx$$

$$g(-t) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xt} dx \quad \text{Replaced } t \text{ by } -t$$

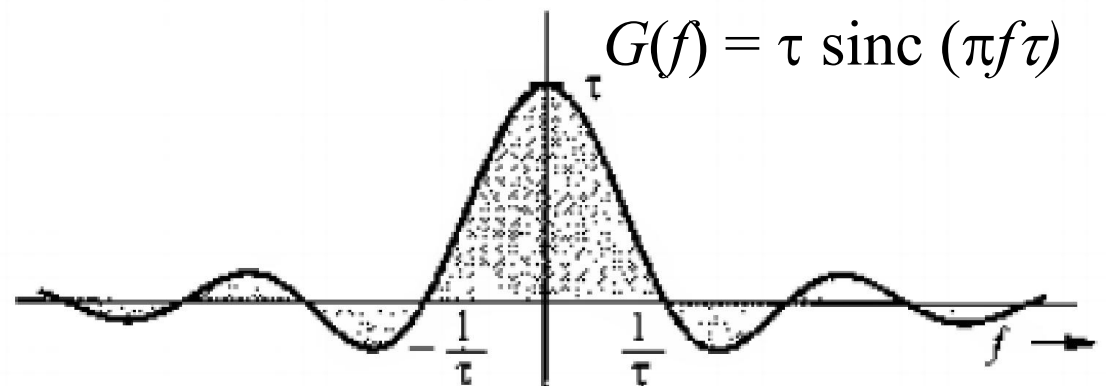
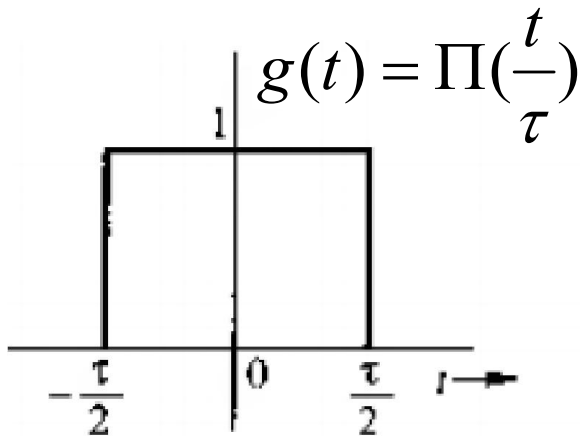
$$g(-f) = \int_{-\infty}^{\infty} G(x) e^{-j2\pi xf} dx \quad \text{Replaced } t \text{ by } f$$

$$g(-f) = \int_{-\infty}^{\infty} G(t) e^{-j2\pi ft} dt \quad \text{Replaced } x \text{ by } t$$

This means, $G(t) \Leftrightarrow g(-f)$

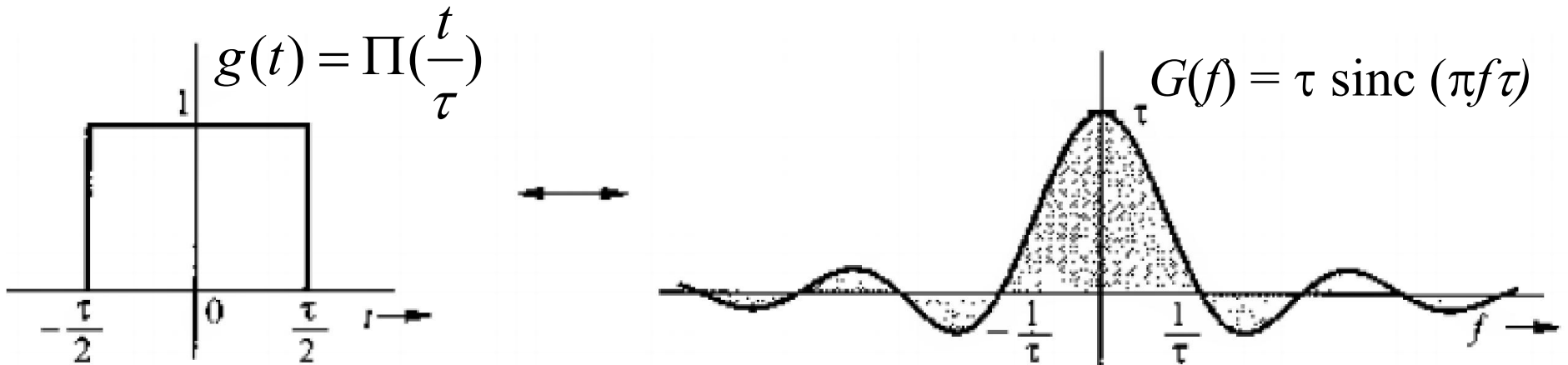
Properties of Fourier Transform

Use of Duality property $G(t) \Leftrightarrow g(-f)$



Properties of Fourier Transform

Use of Duality property $G(t) \Leftrightarrow g(-f)$

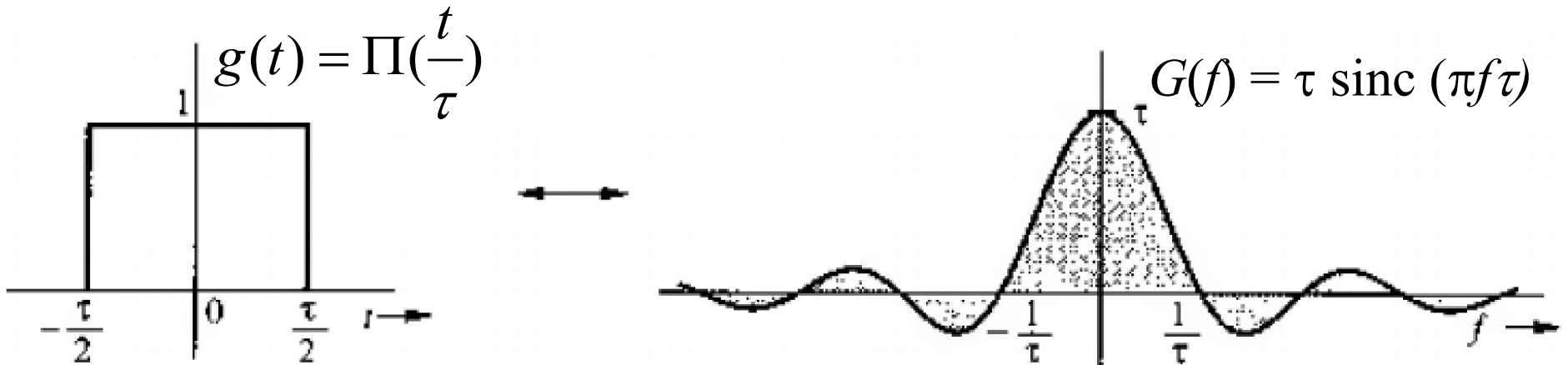


$$G(t) = \tau \operatorname{sinc}(\pi \tau t)$$

$$g(-f) = \Pi\left(-\frac{f}{\tau}\right)$$

Properties of Fourier Transform

Use of Duality property $G(t) \Leftrightarrow g(-f)$



$$G(t) = \tau \operatorname{sinc}(\pi \tau t)$$

$$g(-f) = \Pi\left(-\frac{f}{\tau}\right) = \Pi\left(\frac{f}{\tau}\right)$$

Properties of Fourier Transform

Use of Duality property $G(t) \Leftrightarrow g(-f)$

